

Geocomputation & Machine Learning course

Quick recap

This was a lecture on AWK programming language and its applications in data manipulation, building on previous Bash scripting lessons. Giuseppe demonstrated how AWK can efficiently process large text files line by line, showing examples of pattern matching, column extraction, statistical calculations, and data summarization using the GSIM discharge dataset. Students shared their experiences using Bash for various tasks including climate data analysis, CSV file operations, and downloading data with wget, with Abdul and Sebastián presenting specific use cases they had tried. The lecture covered AWK's syntax with the three operation types (pattern before, action during, and action after reading), variable handling, and integration with Bash commands. Giuseppe showed practical examples including filtering data by location, calculating statistics, creating maps with newplot, and handling missing values, emphasizing how AWK excels at processing well-structured text data before importing it into Python or R. Students were encouraged to practice with their own datasets and explore the powerful combination of Bash and AWK for text data manipulation, with the next lecture scheduled to cover image manipulation using GDAL.

Next steps

Abdul

- Try using AWK for cumulative rainfall analysis on precipitation data.

Annabell

- Use the GSIM dataset for the suggested assignment to practice AWK and Bash scripting.

Areeba

- Explore using pkstat and pktools for histogram and scatterplot analysis between NDVI and

evapotranspiration TIFF files.

Sebastián

- Continue manipulating the CSV file of methane emissions using AWK, exploring more complex operations.

Shiva

- Experiment with AWK on NDVI data, potentially using metadata or summary statistics.

Collaboration

- Sun,: Prepare and upload water chemistry data (CSV/TXT) for future AWK exercises.
- All attendees: Obtain or prepare a CSV/TXT dataset to practice AWK operations demonstrated in the lecture.
- All attendees: Attempt the suggested assignment using the GSIM dataset or their own dataset to rebuild a table with station ID, latitude, longitude, and mean value.
- All attendees: Explore the webpage resource on transforming a Bash for-loop into a multi-core for-loop for parallel processing.

Summary

AWK Text Processing Training Session

Giuseppe led a session on using AWK, a powerful text processing tool, building on previous discussions about Bash. Participants, including Abdul, Sebastián, and Leanda, shared their experiences using Bash for tasks like data manipulation, file downloads, and handling CSV files. Giuseppe explained AWK's efficiency in handling large datasets and demonstrated its use for tasks such as filtering, summarizing, and checking data consistency across multiple files. The session covered AWK's syntax, including pattern matching, column manipulation, and the use of predefined variables like \$1, \$2, and NF. Giuseppe also introduced JupyterLab as a tool for running and experimenting with AWK scripts, emphasizing its utility for data preprocessing before importing into Python or R.

AWK Data Analysis Techniques

Giuseppe demonstrated how to use AWK for text and CSV file analysis, showing examples of cumulative rainfall calculations, statistical operations, and data manipulation techniques including filtering, summing, and conditional operations. Abdul shared his work using Excel and AWK for precipitation analysis, while Sebastián presented a methane emissions dataset from 2022 that he had successfully filtered and summed using AWK. The group discussed practical applications of AWK for handling large datasets and performing statistical analyses, with participants encouraged to practice with their own CSV files during the break.

AWK and Bash Data Manipulation

Giuseppe demonstrated text file manipulation techniques using AWK and Bash commands on a large dataset of 35,000 stations, showing how to extract and visualize latitude/longitude coordinates, count observations, and analyze data patterns. He explained how to use tools like New Plot for fast data visualization and PKtools for image histogram analysis. The session included a suggested assignment for participants to practice manipulating the GSIM dataset and compare Python versus Bash approaches for similar tasks.